
EECS 16A Foundations of Signals, Systems, & Information Processing

Fall 2025 Discussion 4B

1. Intro to DTFS

This is not a question, but a brief intro/overview of DTFS.

Consider a discrete-time p -periodic signal $x : \mathbb{Z} \rightarrow \mathbb{R}$, where

$$\forall n \in \mathbb{Z} \quad x[n+p] = x[n]$$

for some *nonzero* integer p . If p is the smallest *positive* integer for which the equality holds, we refer to it as the *fundamental period* of the signal x .

Such a signal can be expressed as a linear combination of complex exponentials in the form shown by the *DTFS Synthesis Equation* below:

$$\forall n \in \mathbb{Z}, \quad x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n},$$

where $\omega_0 = 2\pi/p$ denotes the *fundamental frequency* of the signal x (if p is the fundamental period), and the set of p coefficients X_k —called the *DTFS coefficients* of x —are unique.

Typically, the index k ranges from 0 to $p-1$. However, depending on convenience, it may range over any set $\langle p \rangle$ of p consecutive integers, such as $[1, p]$ or $[-1, p-2]$, etc.

The DTFS coefficients describe the signal in the frequency domain. Specifically, the k^{th} coefficient X_k carries information about how much the frequency $k\omega_0$ contributes to the signal x —the “amount,” if you will, that each frequency-component $e^{ik\omega_0 n}$ is present in the signal.

The coefficients X_k are given by the *DTFS Analysis Equation*:

$$X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}, \quad k = 0, 1, \dots, p-1.$$

Note that the DT signal may also be complex-valued, and the DTFS synthesis and analysis equations still hold.

2. DTFS Expansion

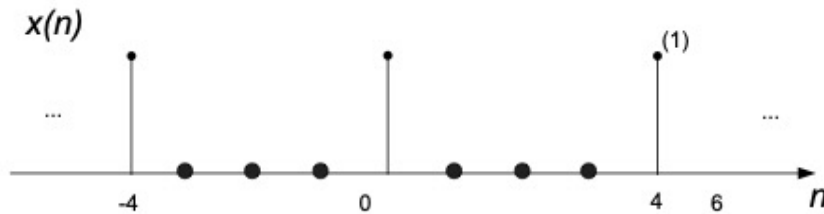
Determine the complex-exponential discrete-time Fourier series (DTFS) expansion for the signal $x : \mathbb{Z} \rightarrow \mathbb{R}$ described below, or explain why no such expansion exists. For the case where a DTFS expansion exists, be sure to identify the period p and the fundamental frequency ω_0 .

(a) In this problem, we introduce the Kronecker-Delta function δ . This function is defined as follows:

$$\delta(n) = \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0 \end{cases}$$

Let $x[n] = \sum_{\ell=-\infty}^{\infty} \delta(n - 4\ell)$ be a signal, for all integers n . To start, provide a well-labeled plot of the signal x first, so you can also see its structure. Then, perform the DTFS expansion, if it exists.

Answer: This is a periodic impulse train.



The signal is periodic with fundamental period $p = 4$, which corresponds to a fundamental frequency of

$$\omega_0 = \frac{2\pi}{4} = \frac{\pi}{2}.$$

The DTFS coefficients are given by

$$X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}, \quad k = 0, 1, 2, 3.$$

But only the $n = 0$ term is nonzero in the summation, so we have

$$X_k = \frac{1}{4} x[0] = \frac{1}{4} \quad k = 0, 1, 2, 3.$$

This result shows that all frequencies $0, \omega_0 = \pi/2, 2\omega_0 = \pi$, and $3\omega_0 = 3\pi/2$ contribute equally to the impulse train. That is,

$$x[n] = \frac{1}{4} \left(1 + e^{i\pi n/2} + e^{i\pi n} + e^{i3\pi n/2} \right).$$

(b) $x[n] = \sin\left(\frac{2\pi}{5}n\right) + \cos\left(\frac{4\pi}{5}n\right), \quad \forall n.$

Answer:

To find the complex-exponential discrete-time Fourier series (DTFS) expansion for $x[n] = \sin\left(\frac{2\pi}{5}n\right) + \cos\left(\frac{4\pi}{5}n\right)$, follow these steps:

1. **Periodicity:** We need to solve for $x[n+p] = x[n]$.

$$x[n+p] = \sin\left(\frac{2\pi}{5}(n+p)\right) + \cos\left(\frac{4\pi}{5}(n+p)\right) \quad (1)$$

$$= \sin\left(\frac{2\pi}{5}n + \frac{2\pi}{5}p\right) + \cos\left(\frac{4\pi}{5}n + \frac{4\pi}{5}p\right) \quad (2)$$

$$= \sin\left(\frac{2\pi}{5}n\right) + \cos\left(\frac{4\pi}{5}n\right) \quad (3)$$

$$= x[n] \quad (4)$$

Therefore, we need $\frac{2\pi}{5}p = 2\pi l$ and $\frac{4\pi}{5}p = 2\pi k$, for integers l, k . However, we note that we can write $2\left(\frac{2\pi}{5}p\right) = 2\pi k$, which is the left hand side of the first equation. So, if we solve for the first equation, we'll also solve the second one.

$$\frac{2\pi}{5}p = 2\pi l \iff p = 5l \implies p = 5$$

by setting $l = 1$. Thus, the signal is periodic with the fundamental period $p = 5$. For completeness, plug $p = 5$ into the second equation to get that:

$$\frac{4\pi}{5}(5) = 2\pi k \iff 4\pi = 2\pi k$$

which is satisfied by the integer $k = 2$.

2. **Fundamental Frequency:** The fundamental angular frequency ω_0 is:

$$\omega_0 = \frac{2\pi}{p} = \frac{2\pi}{5}$$

3. **Finding X_k Coefficients using the Analysis Equation:**

$$X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-i\omega_0 kn}$$

where $p = 5$ and $\omega_0 = \frac{2\pi}{5}$.

To avoid going through tedious computations, recognize that we can apply the inverse Euler's identity to cosine and sine:

- The sine term $\sin\left(\frac{2\pi}{5}n\right)$ can be expressed as $\frac{e^{i\frac{2\pi}{5}n} - e^{-i\frac{2\pi}{5}n}}{2i}$.

- The cosine term $\cos\left(\frac{4\pi}{5}n\right)$ can be expressed as $\frac{e^{i\frac{4\pi}{5}n} + e^{-i\frac{4\pi}{5}n}}{2} = \frac{e^{2i\frac{2\pi}{5}n} + e^{-2i\frac{2\pi}{5}n}}{2}$.

The crucial thing to notice here is that these complex exponentials already have $\omega_0 = \frac{2\pi}{5}$ in their exponents. This means that we can read off the coefficients in front of the complex exponentials as the Fourier coefficients. Thus, combining both terms, we can see that the DTFS coefficients X_k are:

$$X_{-2} = \frac{1}{2}, \quad X_{-1} = -\frac{1}{2i}, \quad X_0 = 0, \quad X_1 = \frac{1}{2i}, \quad X_2 = \frac{1}{2}$$

It makes sense that $X_0 = 0$ since there is no constant offset in the signal, ie. no constant term. Don't forget that the analysis equation has a negative in the exponential. Finally, recall that we can choose any range $\langle p \rangle$ for the index k of the Fourier coefficients. In this case, we chose $\{-2, -1, 0, 1, 2\}$

Thus, the DTFS expansion can be written as:

$$x[n] = \frac{1}{2i}e^{i\frac{2\pi}{5}n} - \frac{1}{2i}e^{-i\frac{2\pi}{5}n} + \frac{1}{2}e^{i\frac{4\pi}{5}n} + \frac{1}{2}e^{-i\frac{4\pi}{5}n}$$

In conclusion, we have:

- Fundamental Period $p = 5$
- Fundamental Frequency $\omega_0 = \frac{2\pi}{5}$
- DTFS Expansion:

$$x[n] = \sum_{k=-2}^2 X_k e^{i\frac{2\pi}{5}kn}$$

where X_k are:

$$X_{-2} = \frac{1}{2}, \quad X_{-1} = -\frac{1}{2i}, \quad X_0 = 0, \quad X_1 = \frac{1}{2i}, \quad X_2 = \frac{1}{2}$$

3. Proving Cool DTFS Properties

Given a periodic signal x with period p and frequency $\omega_0 = \frac{2\pi}{p}$, recall the two golden equations for DTFS:

- Synthesis equation:

$$x[n] = \sum_{k \in \langle p \rangle} X_k e^{ik\omega_0 n}$$

- Analysis equation:

$$X_k = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-ik\omega_0 n}$$

Prove the following properties by "mathematically manipulating" these equations:

- Prove that $X_k = X_{-k}^*$ if and only if x is a real-valued signal.

*Note: this is often called the **Conjugate Symmetry** property.*

Answer: We are proving an if and only if (sometimes denoted as "iff") claim, thus we must prove this from both directions!

We first prove the "if" direction (ie. the backwards direction): assume that x is a real-valued signal. We want to show that $X_k = X_{-k}^*$. We start from the analysis equation to find the conjugate of X_{-k} :

$$\begin{aligned} X_k &= \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-ik\omega_0 n} \\ X_{-k}^* &= \left(\frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-i(-k)\omega_0 n} \right)^* \\ &= \frac{1}{p} \sum_{n \in \langle p \rangle} x^*[n] \left(e^{ik\omega_0 n} \right)^* \\ &= \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-ik\omega_0 n} \\ &= X_k \end{aligned}$$

Next, we prove the "only if" direction (ie. the forwards direction): assume that $X_k = X_{-k}^*$. We want to show that x is a real-valued signal. We proceed using the analysis equation again:

$$\begin{aligned} X_k &= X_{-k}^* \\ \frac{1}{p} \sum_{n \in \langle p \rangle} x(n) e^{-ik\omega_0 n} &= \left(\frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-i(-k)\omega_0 n} \right)^* \\ \frac{1}{p} \sum_{n \in \langle p \rangle} x(n) e^{-ik\omega_0 n} &= \frac{1}{p} \sum_{n \in \langle p \rangle} x^*[n] \left(e^{ik\omega_0 n} \right)^* \\ \sum_{n \in \langle p \rangle} x[n] e^{-ik\omega_0 n} &= \sum_{n \in \langle p \rangle} x^*[n] e^{-ik\omega_0 n} \end{aligned}$$

The corresponding terms on the LHS and RHS of the equation must be equal since the equation holds true for all integers n , so $x[n] = x^*[n]$.

- ii. Let x be a discrete-time periodic signal with DTFS coefficients X_k . Let $y[n] = x[n - n_0]$ be a time-shifted version of x with DTFS coefficients Y_k . Prove that $Y_k = e^{-ik\omega_0 n_0} X_k$.

*Note: this is often called the **Time Shift** property.*

Answer: Observe that $y[n] = x[n - n_0]$ is just a time-shifted version of the signal x , where n_0 is some integer. We begin by expressing the DTFS coefficients Y_k of y using the analysis equation:

$$\begin{aligned} Y_k &= \frac{1}{p} \sum_{n \in \langle p \rangle} y[n] e^{-ik\omega_0 n} \\ &= \frac{1}{p} \sum_{n \in \langle p \rangle} x[n - n_0] e^{-ik\omega_0 n} \end{aligned}$$

Next we perform a change of variables. Let $m = n - n_0$, so $n = m + n_0$. When n runs over one period $\langle p \rangle$, m also runs over one period $\langle p \rangle$, so the summation bounds still hold. Then:

$$\begin{aligned} Y_k &= \frac{1}{p} \sum_{m \in \langle p \rangle} x[m] e^{-ik\omega_0(m+n_0)} \\ &= \frac{1}{p} \sum_{m \in \langle p \rangle} x[m] e^{-ik\omega_0 m} e^{-ik\omega_0 n_0} \\ &= e^{-ik\omega_0 n_0} \left(\frac{1}{p} \sum_{m \in \langle p \rangle} x[m] e^{-ik\omega_0 m} \right) \\ &= e^{-ik\omega_0 n_0} X_k \end{aligned}$$

In the last line, we make a substitution using $X_k = \frac{1}{p} \sum_{m \in \langle p \rangle} x[m] e^{-ik\omega_0 m}$.

The factor $e^{-ik\omega_0 n_0}$ represents a phase-shift of the DTFS coefficient X_k . This shows that a time-shift in the discrete-time signal results in a phase-shift of its DTFS coefficients!